

Seasonal Crop Yield Analytics Dashboard Report

Project Summary

This project presents an end-to-end cloud analytics solution that examines the impact of environmental and agricultural factors on crop productivity. The pipeline uses Amazon S3 for data storage, Snowflake as the cloud data warehouse for data ingestion and transformation, and Power BI for interactive visualization and business intelligence.

The dashboard enables users to explore seasonal trends and identify relationships between rainfall, temperature, humidity, irrigation methods, soil types, and crop yield.

Objectives

- Analyse seasonal agricultural data using cloud-based analytics.
- Build an automated data pipeline from Amazon S3 to Snowflake.
- Transform and query data using SQL.
- Develop interactive dashboards in Power BI.
- Support data-driven agricultural decision-making.

Dashboard Overview

The dashboard consists of four analytical pages:

Rainfall Analysis

- Visualizes rainfall distribution across seasons.
- Compares rainfall levels with crop productivity.
- Highlights seasonal rainfall patterns.

Temperature Analysis

- Examines temperature variations across growing seasons.
- Identifies temperature ranges associated with higher crop yields.

Humidity Analysis

- Analyses humidity trends by season.
- Explores the relationship between humidity and crop performance.

Yield Analysis

- Summarizes crop yield across different crops and seasons.
- Compares productivity under different irrigation methods and soil types.